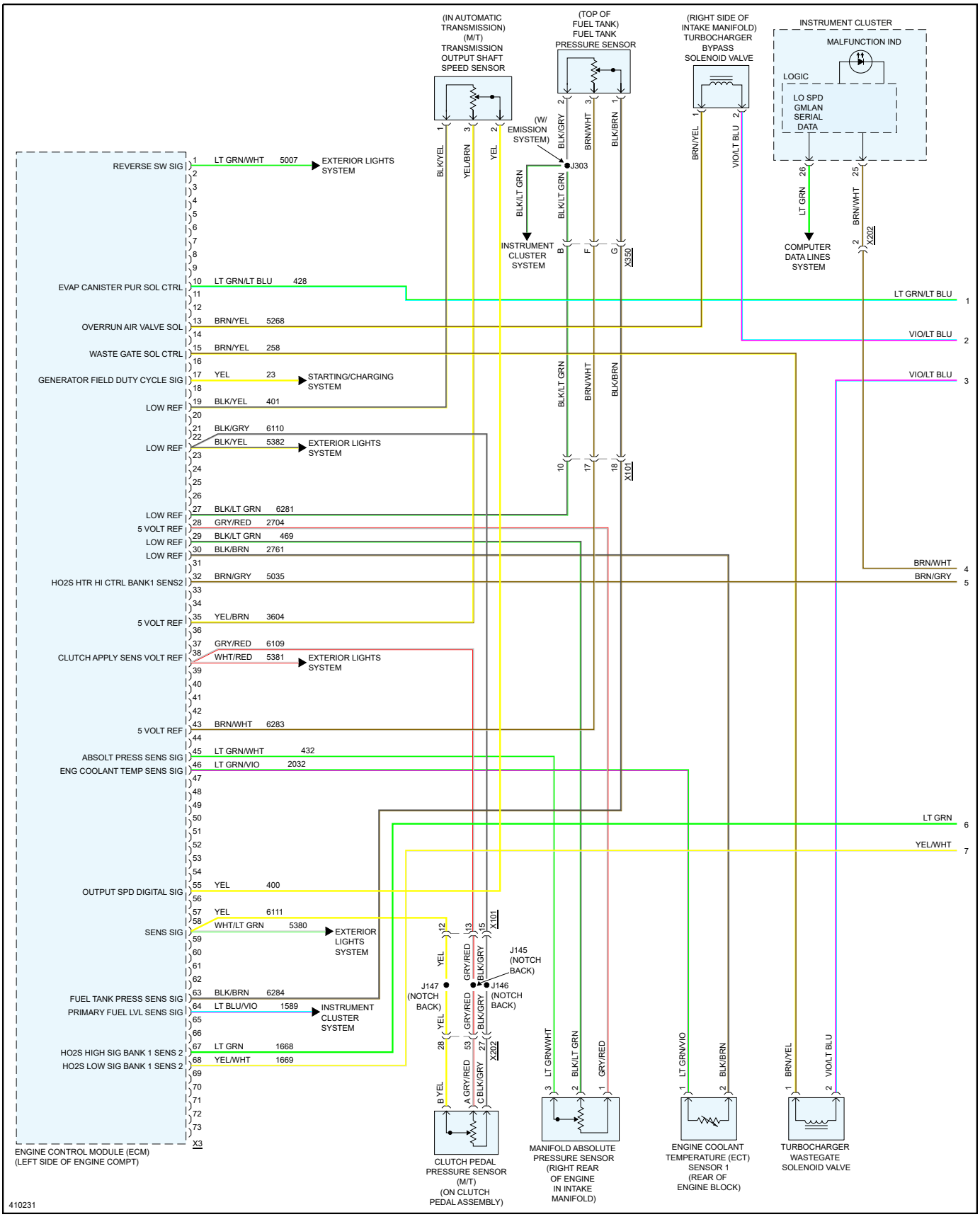
2013 Chevrolet Sonic L4-1.4L Turbo

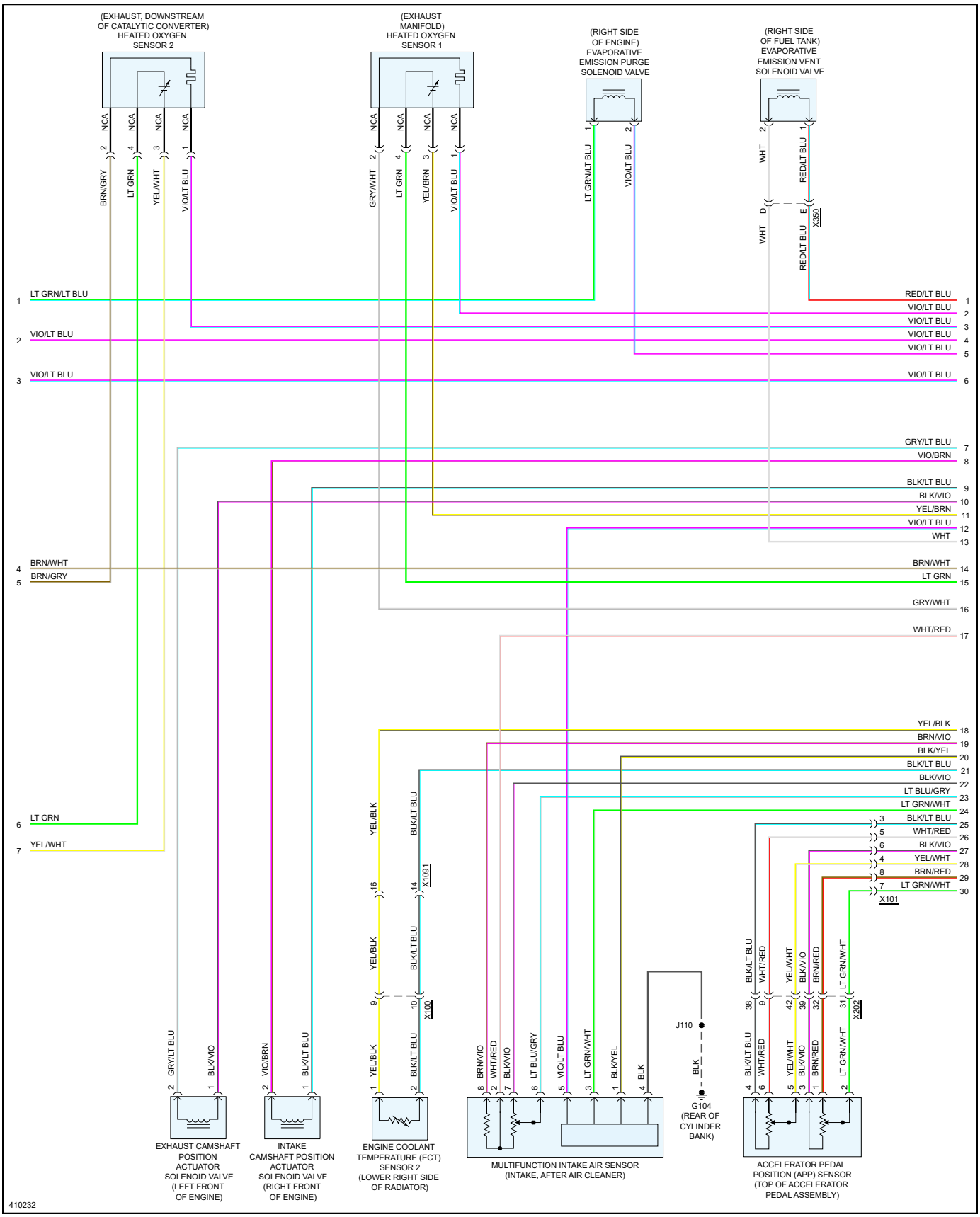
Vehicle > Powertrain Management > Diagrams > Electrical - Interactive Color (Non OE)

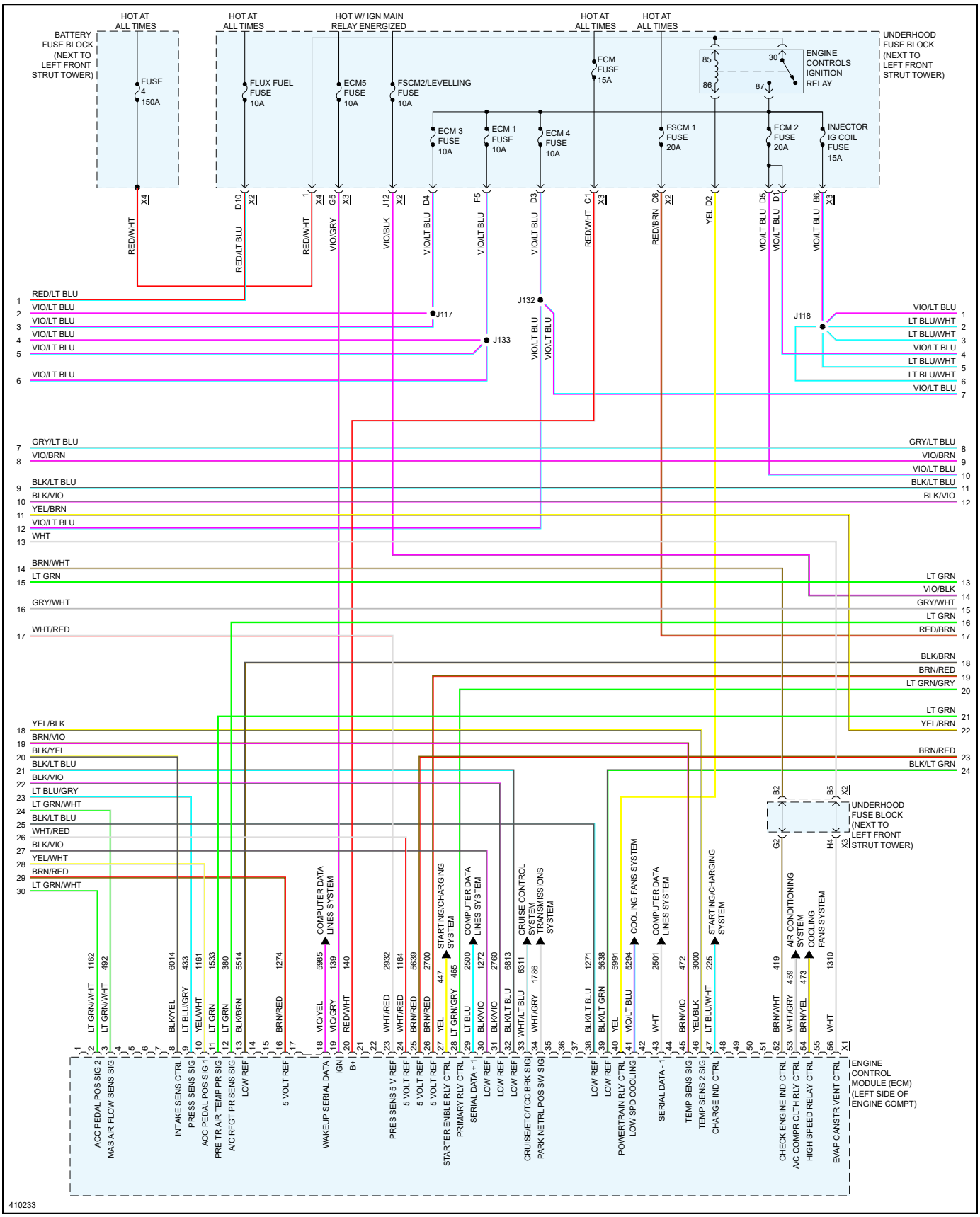
ENGINE CONTROLS

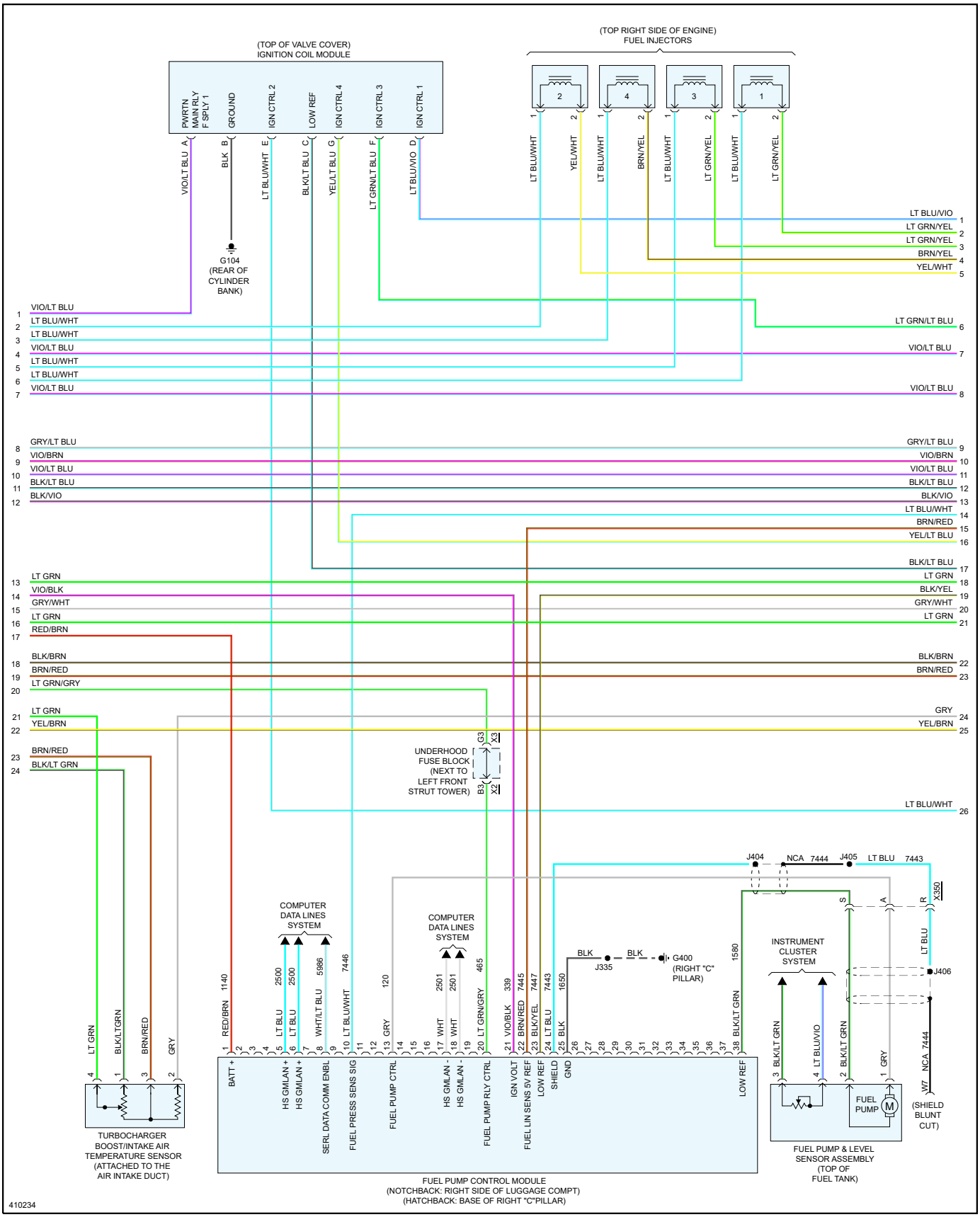
Engine Controls

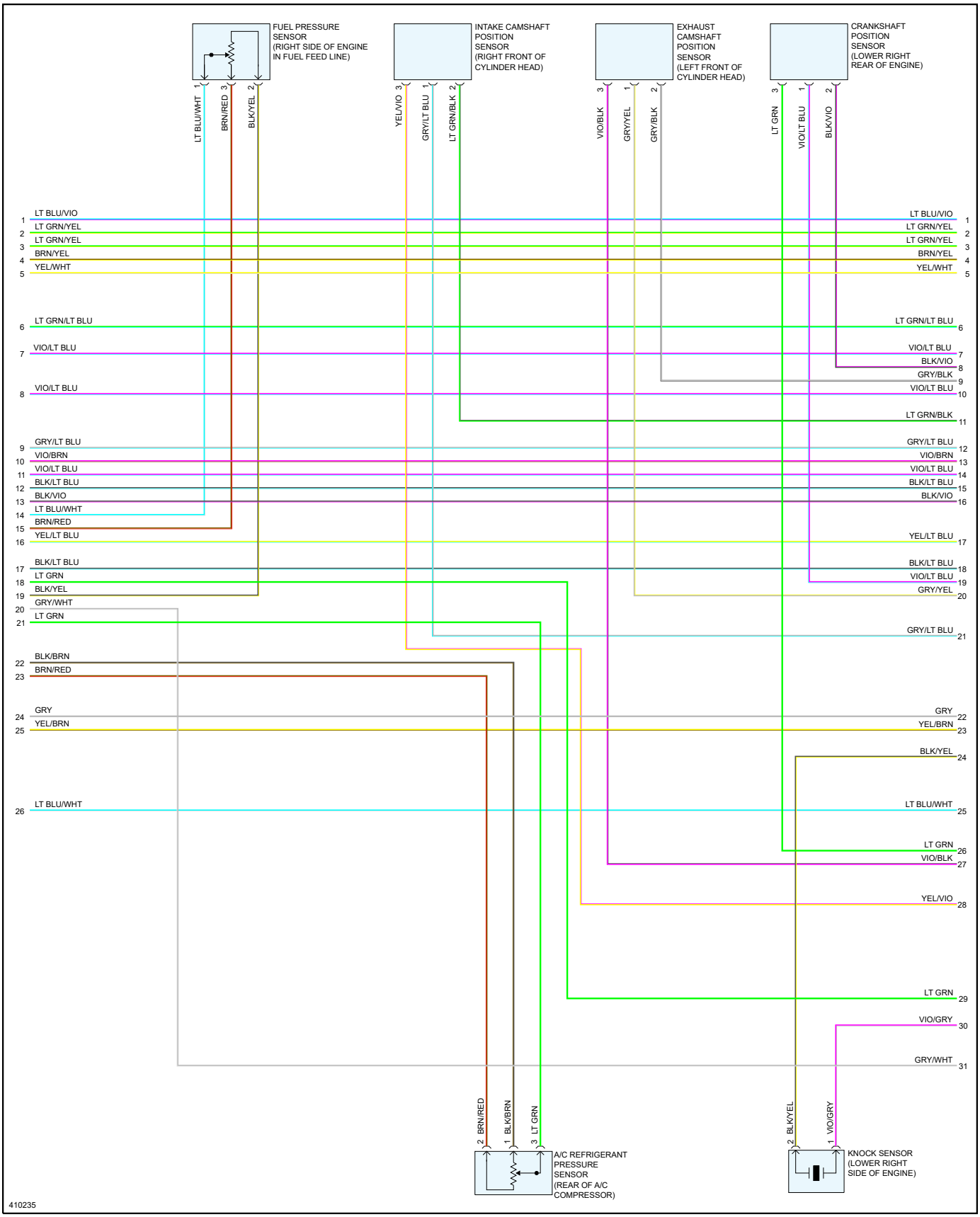
Page 1 of 6

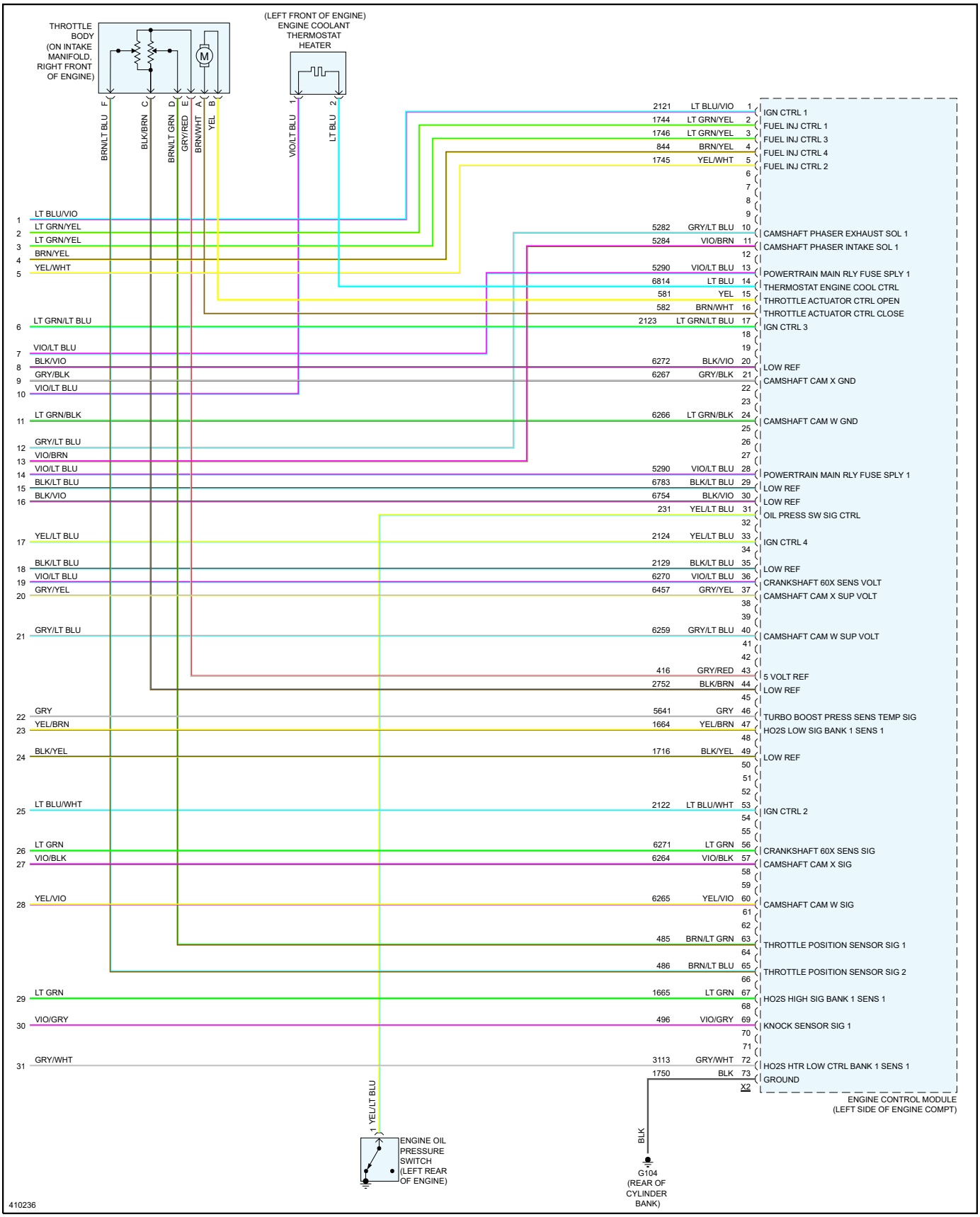

(IN AUTOMATIC TRANSMISSION) (M/T) TRANSMISSION OUTPUT SHAFT SPEED SENSOR
(TOP OF FUEL TANK) FUEL TANK PRESSURE SENSOR
(RIGHT SIDE OF INTAKE MANIFOLD) TURBOCHARGER BYPASS SOLENOID VALVE
INSTRUMENT CLUSTER
MALFUNCTION IND
LOGIC
LO SPD GMLAN SERIAL DATA
(W/ EMISSION SYSTEM)
J303
INSTRUMENT CLUSTER SYSTEM
COMPUTER DATA LINES SYSTEM
REVERSE SW SIG
EVAP CANISTER PUR SOL CTRL
OVERRUN AIR VALVE SOL
WASTE GATE SOL CTRL
GENERATOR FIELD DUTY CYCLE SIG
LOW REF
5 VOLT REF
CLUTCH APPLY SENS VOLT REF
HO2S HTR HI CTRL BANK1 SENS2
ABSOLT PRESS SENS SIG
ENG COOLANT TEMP SENS SIG
OUTPUT SPD DIGITAL SIG
SENS SIG
FUEL TANK PRESS SENS SIG
PRIMARY FUEL LVL SENS SIG
HO2S HIGH SIG BANK 1 SENS 2
HO2S LOW SIG BANK 1 SENS 2
EXTERIOR LIGHTS SYSTEM
STARTING/CHARGING SYSTEM
INSTRUMENT CLUSTER SYSTEM
J145 (NOTCH BACK)
J146 (NOTCH BACK)
J147 (NOTCH BACK)
ENGINE CONTROL MODULE (ECM) (LEFT SIDE OF ENGINE COMPT)
CLUTCH PEDAL PRESSURE SENSOR (M/T) (ON CLUTCH PEDAL ASSEMBLY)
MANIFOLD ABSOLUTE PRESSURE SENSOR (RIGHT REAR OF ENGINE IN INTAKE MANIFOLD)
ENGINE COOLANT TEMPERATURE (ECT) SENSOR 1 (REAR OF ENGINE BLOCK)
TURBOCHARGER WASTEGATE SOLENOID VALVE
410231


(EXHAUST, DOWNSTREAM OF CATALYTIC CONVERTER) HEATED OXYGEN SENSOR 2
(EXHAUST MANIFOLD) HEATED OXYGEN SENSOR 1
(RIGHT SIDE OF ENGINE) EVAPORATIVE EMISSION PURGE SOLENOID VALVE
(RIGHT SIDE OF FUEL TANK) EVAPORATIVE EMISSION VENT SOLENOID VALVE
NCA
BRN/GRY
LT GRN
YEL/WHT
VIO/LT BLU
GRY/WHT
LT GRN/LT BLU
WHT
RED/LT BLU
X350
LT GRN/LT BLU
VIO/LT BLU
BRN/WHT
BRN/GRY
GRY/LT BLU
VIO/BRN
BLK/LT BLU
BLK/VIO
YEL/BRN
WHT
LT GRN
YEL/WHT
GRY/WHT
WHT/RED
YEL/BLK
BRN/VIO
BLK/YEL
LT BLU/GRY
LT GRN/WHT
X1091
X100
X101
X202
J110
G104
(REAR OF CYLINDER BANK)
EXHAUST CAMSHAFT POSITION ACTUATOR SOLENOID VALVE (LEFT FRONT OF ENGINE)
INTAKE CAMSHAFT POSITION ACTUATOR SOLENOID VALVE (RIGHT FRONT OF ENGINE)
ENGINE COOLANT TEMPERATURE (ECT) SENSOR 2 (LOWER RIGHT SIDE OF RADIATOR)
MULTIFUNCTION INTAKE AIR SENSOR (INTAKE, AFTER AIR CLEANER)
ACCELERATOR PEDAL POSITION (APP) SENSOR (TOP OF ACCELERATOR PEDAL ASSEMBLY)
410232


HOT AT ALL TIMES
HOT AT ALL TIMES
HOT W/ IGN MAIN RELAY ENERGIZED
HOT AT ALL TIMES
HOT AT ALL TIMES
BATTERY FUSE BLOCK (NEXT TO LEFT FRONT STRUT TOWER)
FUSE 4 150A
FLUX FUEL FUSE 10A
ECM5 FUSE 10A
FSCM2/LEVELLING FUSE 10A
ECM FUSE 15A
ENGINE CONTROLS IGNITION RELAY
UNDERHOOD FUSE BLOCK (NEXT TO LEFT FRONT STRUT TOWER)
ECM 3 FUSE 10A
ECM 1 FUSE 10A
ECM 4 FUSE 10A
FSCM 1 FUSE 20A
ECM 2 FUSE 20A
INJECTOR IG COIL FUSE 15A
J117
J132
J133
J118
UNDERHOOD FUSE BLOCK (NEXT TO LEFT FRONT STRUT TOWER)
ENGINE CONTROL MODULE (ECM) (LEFT SIDE OF ENGINE COMPT)
ACC PEDAL POS SIG 2
MAS AIR FLOW SENS SIG
INTAKE SENS CTRL
PRESS SENS SIG
ACC PEDAL POS SIG 1
PRE TR AIR TEMP PR SIG
A/C RFGT PR SENS SIG
LOW REF
5 VOLT REF
WAKEUP SERIAL DATA
IGN
B+
PRES SENS 5 V REF
5 VOLT REF
5 VOLT REF
5 VOLT REF
STARTER ENBLE RLY CTRL
PRIMARY RLY CTRL
SERIAL DATA + 1
LOW REF
LOW REF
LOW REF
CRUISE/ETC/TCC BRK SIG
PARK NETRL POS SW SIG
LOW REF
LOW REF
POWERTRAIN RLY CTRL
LOW SPD COOLING
SERIAL DATA - 1
TEMP SENS SIG
TEMP SENS 2 SIG
CHARGE IND CTRL
CHECK ENGINE IND CTRL
A/C COMPR CLTH RLY CTRL
HIGH SPEED RELAY CTRL
EVAP CANSTR VENT CTRL
COMPUTER DATA LINES SYSTEM
STARTING/CHARGING SYSTEM
COMPUTER DATA LINES SYSTEM
CRUISE CONTROL SYSTEM
TRANSMISSIONS SYSTEM
COOLING FANS SYSTEM
COMPUTER DATA LINES SYSTEM
STARTING/CHARGING SYSTEM
AIR CONDITIONING SYSTEM
COOLING FANS SYSTEM
410233


(TOP OF VALVE COVER)
IGNITION COIL MODULE
PWRTN MAIN RLY F SPLY 1
GROUND
IGN CTRL 2
LOW REF
IGN CTRL 4
IGN CTRL 3
IGN CTRL 1
(TOP RIGHT SIDE OF ENGINE)
FUEL INJECTORS
G104
(REAR OF CYLINDER BANK)
UNDERHOOD FUSE BLOCK
(NEXT TO LEFT FRONT STRUT TOWER)
J404
NCA 7444
J405
LT BLU 7443
J335
G400
(RIGHT "C" PILLAR)
J406
INSTRUMENT CLUSTER SYSTEM
COMPUTER DATA LINES SYSTEM
COMPUTER DATA LINES SYSTEM
TURBOCHARGER BOOST/INTAKE AIR TEMPERATURE SENSOR (ATTACHED TO THE AIR INTAKE DUCT)
FUEL PUMP CONTROL MODULE
(NOTCHBACK: RIGHT SIDE OF LUGGAGE COMPT)
(HATCHBACK: BASE OF RIGHT "C"PILLAR)
FUEL PUMP & LEVEL SENSOR ASSEMBLY (TOP OF FUEL TANK)
FUEL PUMP
(SHIELD BLUNT CUT)
BATT +
HS GMLAN +
HS GMLAN +
SERL DATA COMM ENBL
FUEL PRESS SENS SIG
FUEL PUMP CTRL
HS GMLAN -
HS GMLAN -
FUEL PUMP RLY CTRL
IGN VOLT
FUEL LIN SENS 5V REF
LOW REF
SHIELD
GND
LOW REF
410234


FUEL PRESSURE SENSOR (RIGHT SIDE OF ENGINE IN FUEL FEED LINE)
INTAKE CAMSHAFT POSITION SENSOR (RIGHT FRONT OF CYLINDER HEAD)
EXHAUST CAMSHAFT POSITION SENSOR (LEFT FRONT OF CYLINDER HEAD)
CRANKSHAFT POSITION SENSOR (LOWER RIGHT REAR OF ENGINE)
LT BLU/WHT
BRN/RED
BLK/YEL
YEL/VIO
GRY/LT BLU
LT GRN/BLK
VIO/BLK
GRY/YEL
GRY/BLK
LT GRN
VIO/LT BLU
BLK/VIO
LT BLU/VIO
LT GRN/YEL
LT GRN/YEL
BRN/YEL
YEL/WHT
LT GRN/LT BLU
VIO/LT BLU
GRY/BLK
VIO/LT BLU
LT GRN/BLK
GRY/LT BLU
VIO/BRN
VIO/LT BLU
BLK/LT BLU
BLK/VIO
LT BLU/WHT
BRN/RED
YEL/LT BLU
BLK/LT BLU
LT GRN
BLK/YEL
GRY/WHT
LT GRN
GRY/LT BLU
BLK/BRN
BRN/RED
GRY
YEL/BRN
BLK/YEL
LT BLU/WHT
LT GRN
VIO/BLK
YEL/VIO
LT GRN
VIO/GRY
GRY/WHT
BRN/RED
BLK/BRN
LT GRN
A/C REFRIGERANT PRESSURE SENSOR (REAR OF A/C COMPRESSOR)
BLK/YEL
VIO/GRY
KNOCK SENSOR (LOWER RIGHT SIDE OF ENGINE)
410235


THROTTLE BODY (ON INTAKE MANIFOLD, RIGHT FRONT OF ENGINE)
(LEFT FRONT OF ENGINE) ENGINE COOLANT THERMOSTAT HEATER
F
C
D
E
A
B
1
2
BRN/LT BLU
BLK/BRN
BRN/LT GRN
GRY/RED
BRN/WHT
YEL
VIO/LT BLU
LT BLU
2121 LT BLU/VIO 1 IGN CTRL 1
1744 LT GRN/YEL 2 FUEL INJ CTRL 1
1746 LT GRN/YEL 3 FUEL INJ CTRL 3
844 BRN/YEL 4 FUEL INJ CTRL 4
1745 YEL/WHT 5 FUEL INJ CTRL 2
6
7
8
9
5282 GRY/LT BLU 10 CAMSHAFT PHASER EXHAUST SOL 1
5284 VIO/BRN 11 CAMSHAFT PHASER INTAKE SOL 1
12
5290 VIO/LT BLU 13 POWERTRAIN MAIN RLY FUSE SPLY 1
6814 LT BLU 14 THERMOSTAT ENGINE COOL CTRL
581 YEL 15 THROTTLE ACTUATOR CTRL OPEN
582 BRN/WHT 16 THROTTLE ACTUATOR CTRL CLOSE
2123 LT GRN/LT BLU 17 IGN CTRL 3
18
19
6272 BLK/VIO 20 LOW REF
6267 GRY/BLK 21 CAMSHAFT CAM X GND
22
23
6266 LT GRN/BLK 24 CAMSHAFT CAM W GND
25
26
27
5290 VIO/LT BLU 28 POWERTRAIN MAIN RLY FUSE SPLY 1
6783 BLK/LT BLU 29 LOW REF
6754 BLK/VIO 30 LOW REF
231 YEL/LT BLU 31 OIL PRESS SW SIG CTRL
32
2124 YEL/LT BLU 33 IGN CTRL 4
34
2129 BLK/LT BLU 35 LOW REF
6270 VIO/LT BLU 36 CRANKSHAFT 60X SENS VOLT
6457 GRY/YEL 37 CAMSHAFT CAM X SUP VOLT
38
39
6259 GRY/LT BLU 40 CAMSHAFT CAM W SUP VOLT
41
42
416 GRY/RED 43 5 VOLT REF
2752 BLK/BRN 44 LOW REF
45
5641 GRY 46 TURBO BOOST PRESS SENS TEMP SIG
1664 YEL/BRN 47 HO2S LOW SIG BANK 1 SENS 1
48
1716 BLK/YEL 49 LOW REF
50
51
52
2122 LT BLU/WHT 53 IGN CTRL 2
54
55
6271 LT GRN 56 CRANKSHAFT 60X SENS SIG
6264 VIO/BLK 57 CAMSHAFT CAM X SIG
58
59
6265 YEL/VIO 60 CAMSHAFT CAM W SIG
61
62
485 BRN/LT GRN 63 THROTTLE POSITION SENSOR SIG 1
64
486 BRN/LT BLU 65 THROTTLE POSITION SENSOR SIG 2
66
1665 LT GRN 67 HO2S HIGH SIG BANK 1 SENS 1
68
496 VIO/GRY 69 KNOCK SENSOR SIG 1
70
71
3113 GRY/WHT 72 HO2S HTR LOW CTRL BANK 1 SENS 1
1750 BLK 73 GROUND
X2
ENGINE CONTROL MODULE (LEFT SIDE OF ENGINE COMPT)
1 LT BLU/VIO
2 LT GRN/YEL
3 LT GRN/YEL
4 BRN/YEL
5 YEL/WHT
6 LT GRN/LT BLU
7 VIO/LT BLU
8 BLK/VIO
9 GRY/BLK
10 VIO/LT BLU
11 LT GRN/BLK
12 GRY/LT BLU
13 VIO/BRN
14 VIO/LT BLU
15 BLK/LT BLU
16 BLK/VIO
17 YEL/LT BLU
18 BLK/LT BLU
19 VIO/LT BLU
20 GRY/YEL
21 GRY/LT BLU
22 GRY
23 YEL/BRN
24 BLK/YEL
25 LT BLU/WHT
26 LT GRN
27 VIO/BLK
28 YEL/VIO
29 LT GRN
30 VIO/GRY
31 GRY/WHT
1 YEL/LT BLU
ENGINE OIL PRESSURE SWITCH (LEFT REAR OF ENGINE)
BLK
G104 (REAR OF CYLINDER BANK)
410236